



Rates and Percent Small-Group Lab

Grade 6 • Mathematics
Small-Group Work

Original Practice • Assessment • Complete Teacher Key



Teacher Guide

Level: Grade 6

State: North Carolina

Subject: Mathematics

Resource type: Small-Group Work

Estimated time: 45-55 minutes

Standard / objective: NC.6.RP.3 - Use ratio and rate reasoning to solve real-world and mathematical problems, including unit rates and percentages.

Source note: North Carolina Department of Public Instruction, Mathematics Standards Vertical Progression:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>; NC Check-Ins 2.0 Mathematics

Specifications: <https://www.dpi.nc.gov/documents/accountability/testing/nc-checkin/nc-check-ins-20-mathematics-grades-3-8-specifications> (accessed September 14, 2026).

What is included: branded cover, teacher guide, student directions, 16 original practice items, 5-item exit ticket, complete answer key/worked guidance, differentiation supports, printing notes, and license statement.

Purpose: Use tables, unit rates, and percentages to compare situations and justify decisions.

Preparation: Print student pages single-sided or assign through a secure LMS. Review the key and select tools allowed for your learners. No special commercial curriculum is required.

Access and differentiation

AIG: require a second representation, proof, counterexample, transfer task, or student-created extension. **504:** provide extended time, chunked sections, breaks, reduced copying, and accessible digital zoom as specified by the plan. **EC:** preteach vocabulary, model one item, provide manipulatives/visuals, and accept aligned response modes from the IEP. **ESL:** preview key terms, provide bilingual resources and sentence frames, and allow partner rehearsal. Use each learner's current plan; this guide does not replace it.



Student Practice

Learning focus: Use tables, unit rates, and percentages to compare situations and justify decisions.

Directions: Complete all items. Show calculations, cite evidence, or explain your reasoning when requested. Check your work before submitting.

1. 3 notebooks cost \$7.50. Unit price?
2. A car travels 180 miles in 3 hours. Unit rate?
3. Complete equivalent ratio $4:7 = 12: \underline{\hspace{1cm}}$.
4. 25% of 80
5. A \$40 item is discounted 15%. Discount amount?
6. A recipe uses 2 cups for 5 servings. Cups per serving?
7. Which is cheaper: 6 pens for \$9 or 10 for \$16?
8. 18 is what percent of 60?



Student Practice - Continued

9. Increase 50 by 12%.
10. A map scale is 1 inch:8 miles. 3.5 inches represents?
11. A runner completes 5 km in 25 min. Minutes per km?
12. Explain why 0.35 and 35% are equivalent.
13. Create a ratio table for 4 tickets costing \$30.
14. A class of 24 has 18 present. Percent present?
15. Tax is 7% on \$60. Total?
16. Compare $\frac{7}{10}$, 68%, and 0.72.



Exit Ticket

Complete independently in 8-10 minutes.

1. State the main skill or concept in your own words.
2. Create one accurate example that uses today's skill.
3. Explain how you checked your example.
4. Identify one common error and correct it.
5. Rate your confidence from 1-4 and name your next practice step.

Quick rubric: 4 = accurate, complete, and well justified; 3 = accurate with minor omission; 2 = partial understanding; 1 = limited evidence; 0 = blank or unrelated.



Complete Teacher Answer Key

1. \$2.50
2. 60 mph
3. 21
4. 20
5. \$6
6. 0.4 cup
7. 6 for \$9; \$1.50 each vs. \$1.60
8. 30%
9. 56
10. 28 miles
11. 5
12. Percent means per hundred; $35/100 = 0.35$
13. 1:\$7.50, 2:\$15, 4:\$30
14. 75%
15. \$64.20
16. $68\% < 7/10 < 0.72$

Exit ticket guidance: Responses vary. Award full credit when the student accurately explains the focus, supplies a correct example, describes a valid check, identifies and corrects a realistic misconception, and chooses a specific next step.

Verification: Fixed responses and calculations were checked during production. Open responses are evaluated against the criteria above.

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